Lab 1: Alignment and Geometric Optics

Names: HyunSeo Lee, Jerry Zhang

Lab report due: Midnight, Friday, Oct 2, 2026

Objectives

- Practice basic optical alignment and geometric optics techniques
- Observe the image formation behavior in a finite and infinite conjugate imaging system
- See how lateral, axial, and angular magnification determine the final image formed

Basic Rules

Before you start, remember:

- Do not touch the lens surfaces directly; use a tissue and handle by the edges.
- Do not look directly into the laser or point it at other people!
- When aligning, do not put your eyes on the level of the beam, to avoid accidental reflections into your eye.
- If your laser is sending stray light to other tables, use a book or paper etc. to block the beam.

Lab report: Include answers to bolded questions in your group lab report. Please type question answers; handwritten drawings or calculations are ok, but make sure the final submission is a PDF file.

Basic alignment techniques

Laser alignment: Begin system setup with a straight, level laser beam starting from one end of the board. Alignment is easiest with the beam parallel to breadboard holes. Observe the x and y positions (i.e., on the transverse plane perpendicular to optical axis) of the beam close to the diode output and at the opposite end of the board using the paper target attached to a mounting post (which acts as a vertical height marker) at the same lateral board locations. To describe the beam path as a line, remember you must have two points of reference to guide alignment. Adjust laser using kinematic mount knobs, rotation about post, and/or translation of base until the beam is straight in the horizontal and vertical directions. **Include a picture of your setup/targets in the lab report.**

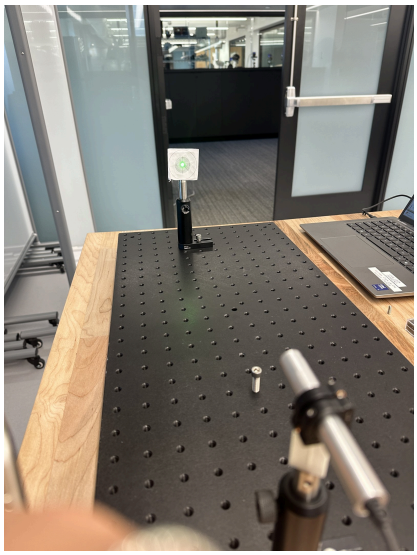
Images of our setup is shown below:



(a) Close-up view of beam location



(b) Target close to laser

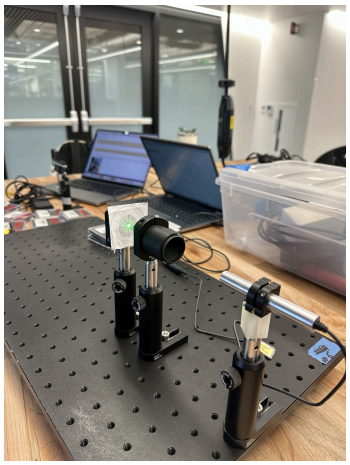


(c) Target far from laser

Lens alignment: Now practice aligning the 30 mm lens in the beam path ~1f + several extra cm from the laser. Note that you may have to correct tilt and position errors iteratively.
Place the paper target about 1f beyond the incoming lens position and note the beam location. Roughly center the mounted lens by observing where the laser falls on the front surface of the lens; does the beam appear in the same location?
For lateral alignment, first move the lens back and forth in the transverse plane perpendicular to the beam. **What changes do you observe in the position of the light on the target? Explain using your knowledge of geometric optics** (Hint: Consider ray tracing for a lens with an 'optical axis' collinear or far from the beam's optical axis). Adjust the lens lateral position so that the beam is located at the pre-lens position.
Tilt error can be corrected by guiding the reflection off of the lens surface back into the path it originated from (Note: reflection may be small depending on source brightness; most lenses we are using are anti-reflection coated. If you are interested in learning more about the surface reflections, check out the Fresnel equations https://www.rp-photonics.com/fresnel_equations.html).

Q: What changes do you observe in the position of the light on the target? Explain using your knowledge of geometric optics

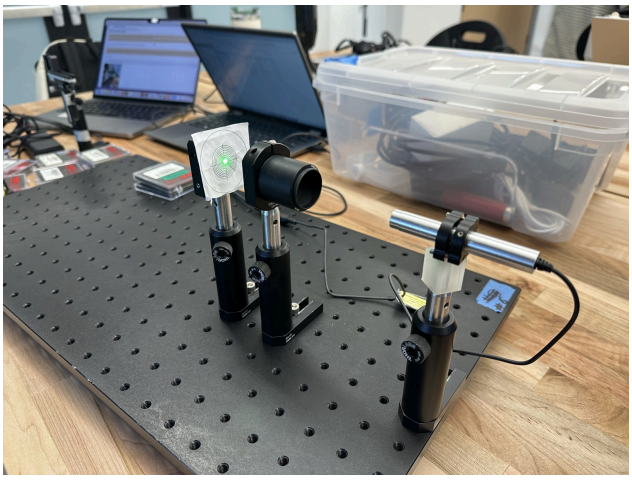
A: The position of the light on the target moves in the same direction as the lens when it is moved laterally. If the beam does not pass through the center of the lens, the lens refracts the beam to shift its position on the target. When the beam passes through the center of the lens, it is not deflected and the spot returns to its original, pre-lens position. We used this knowledge to center the system laterally.



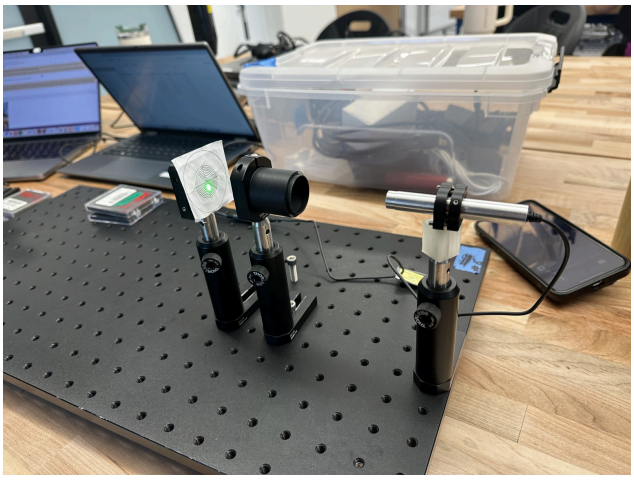
Alignment of 30 mm lens in beam path

Basic geometric optics

Principal rays: In class, three different principal rays for a lens were discussed. Recall these three rays and their relationships to a lens and its focal length. Try to experimentally recreate two of these rays with your laser and 30 mm lens, specifically the ray traveling through the center of a lens and the ray traveling parallel to the optical axis. **Record the result from the two setups.**
NOTE: It will be difficult to ensure that your beam is traveling parallel to the optical axis, especially as the tilt of the lens is difficult to accurately tune. Try to control this to the best of your ability and use the breadboard markings as a reference whenever possible. Further, the interaction of the beam with the single lens will change the beam shape as it propagates. Try to think about the general propagation direction and not the beam shape when trying to observe the principal rays.



(a) Principal ray: ray traveling through center of a lens

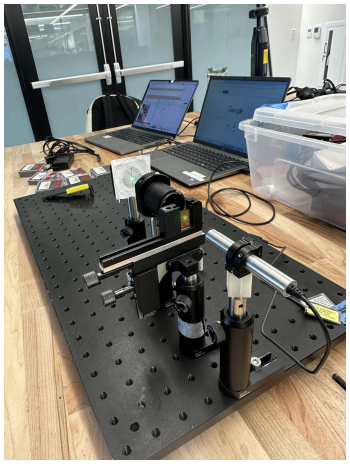


(b) Principal ray: ray traveling parallel to the optical axis

A: For the ray travelling through center of the lens, the beam experiences no change in propagation direction, so the laser spot is at the same position on the target.

For the ray travelling parallel to the optical axis, the lens refract the beam so that it crosses the optical axis at the back focal point, one focal length (~30mm) behind the lens. On the target, as shown in the image above (b), the spot moves from its original height toward the optical axis, reaching the axis when the target is at the focal point, and cross to the opposite side when the target is placed beyond the focal point.

Estimation of focal length: Next you will be estimating the focal length of the lens. For the illumination source, place the laser on a mounting post. Direct the beam along a line of the breadboard. You can confirm this with a mounting post and ensuring the beam is at about the same height and lateral position at multiple different points along the path (see 1a). Now position the R1L1SN1 calibration tool so the laser illuminates the pattern along the breadboard axis. This will be your object. Now place a lens of your choice along the same pathway as the laser. Use a piece of paper (as screen) to find where the image of the object is formed most sharply. Record the distance from the object to the lens, and from the lens to the image. **Use these measurements and the thin lens formula to estimate the focal length of the lens.** Repeat the process for a few different object distances to **find an average focal length for the lens. Include focal length, raw measurements, and your calculation in the report.**



System to determine focal length of lens

The table below shows the measurements we took to obtain the focal length of the lens. (D_o = object distance, D_i = image distance)

D_o (mm)	D_i (mm)	Focal length
56	65	30.0826446
51	78	30.8372093
46	90	30.4411765
40	105	28.9655172
60	55	28.6956522

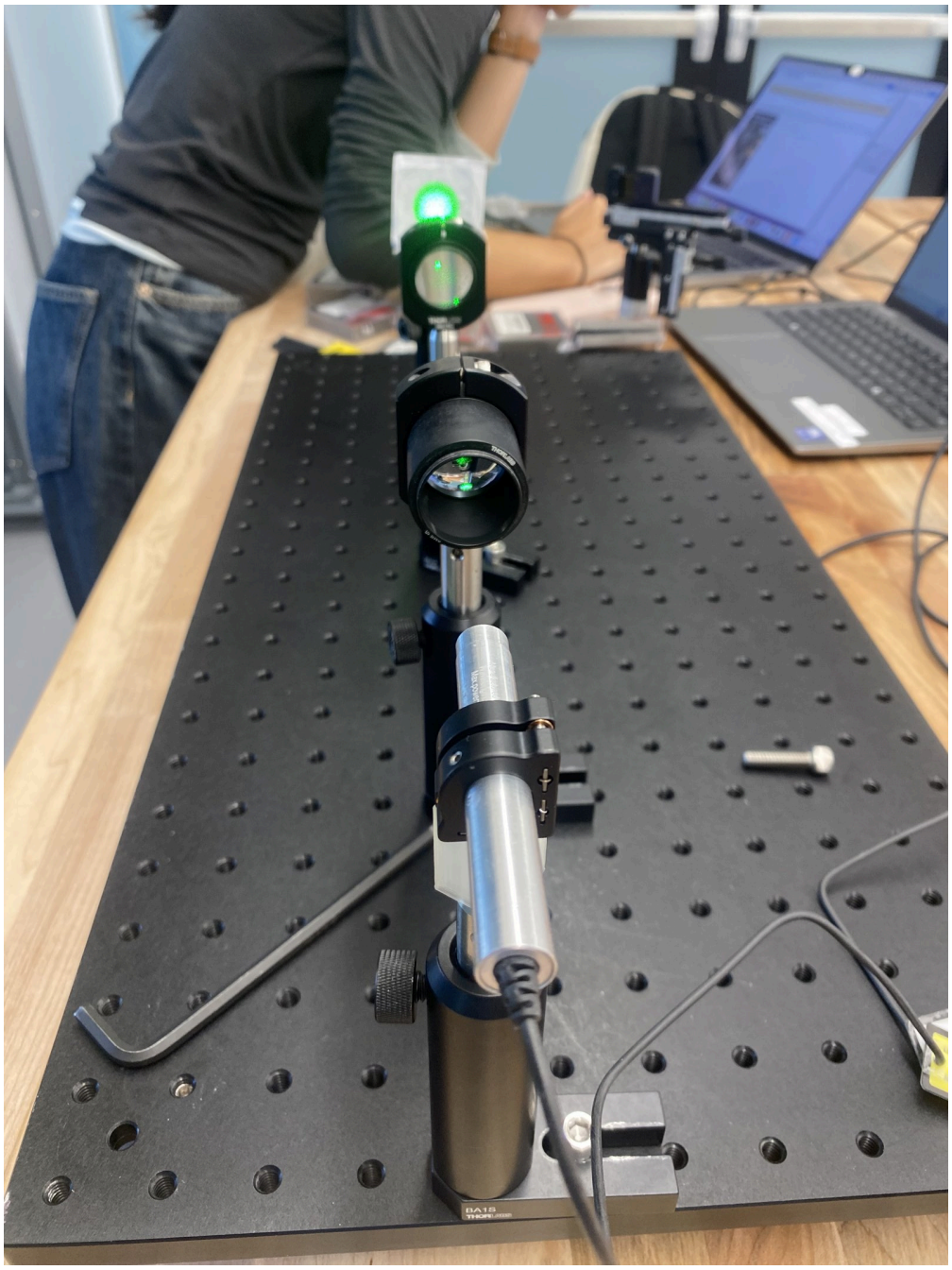
We used the thin lens equation,

$$\frac{1}{f} = \frac{1}{D_o} + \frac{1}{D_i}$$

to calculate the focal length for each pair of object and image distances. We then averaged the five calculated focal lengths, giving an average focal length of approximately **29.8 ± 0.9 mm** (mean ± standard deviation), within 0.7% of the nominal 30 mm. We believe the spread and discrepency comes from human error in judging where the image is sharpest with the human eye and from measuring distances to the center of a lens with a ruler not a more precise and sophisticated system.

Finite and infinite conjugate imaging

Keeping the laser and 30 mm lens aligned, add and align the 150 mm lens in the optical path to set the lenses 2f apart. Note: “2f” means two focal lengths. In this case, this means f1+f2. **What should the output beam look like if the lenses are 2f apart? Verify the beam size at several distances from the second lens.**



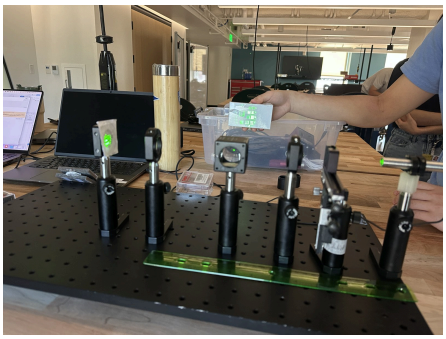
4f system

Answer: If the lenses are 2f apart, the output beam should be collimated. The first lens focuses the collimated beam at its focal point, which is the focal point of the second lens, so the second lens collimates the beam. We measured the beam size at several distances from the second lens, and got consistently 24mm in diameter.

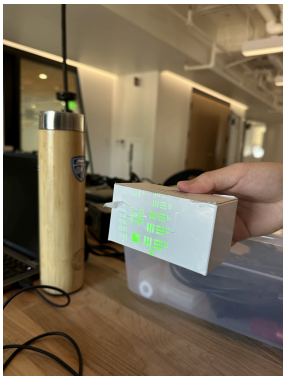
You will now use the alignment of the 2f lenses to create a 4f system. Place the R1L1S1N calibration tool in the optical path so that the laser illuminates the grid pattern. Change the axial position of the tool to try and place it at the focal length of the first lens. Add a mirror or beamsplitter between the first and second lenses to create a long image distance as a reference. Adjust the position between the calibration glass (object) and the first lens while observing the image formed beyond the mirror until you observe the expected pattern that results from alignment at the focal length. Using the lens equation, **calculate the magnification of the overall relay. Compare the magnification with object and image measurements.**



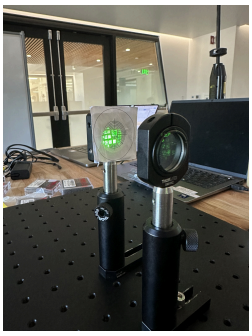
(a) Full 4f system



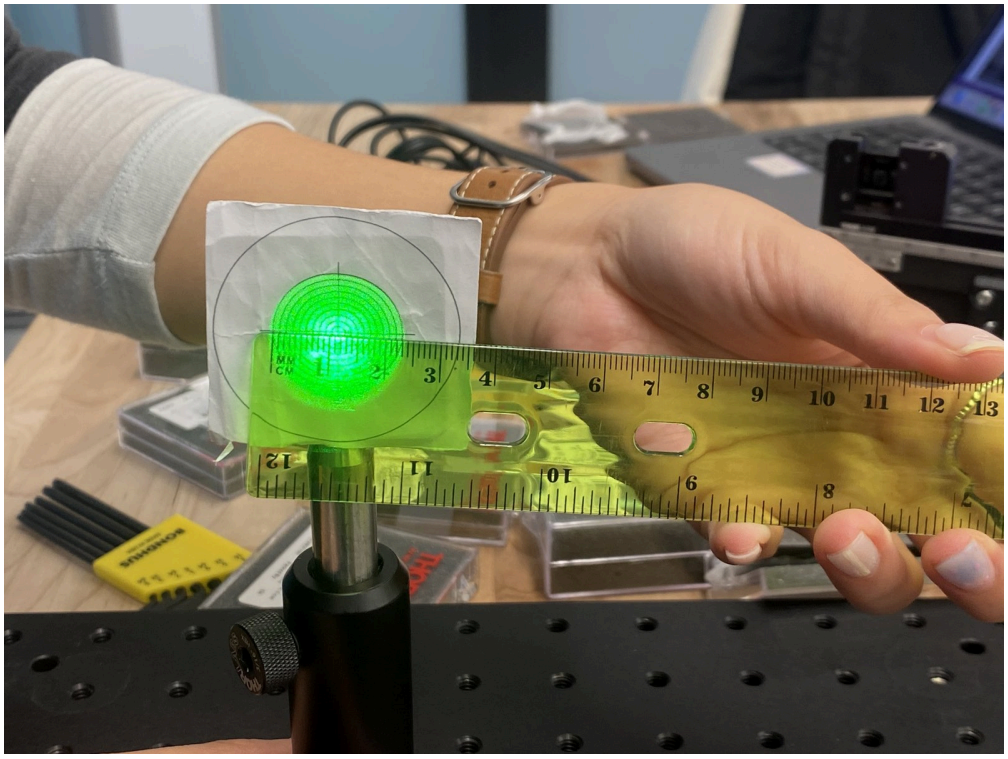
(b) 4f system with intermediate image



(c) Intermediate image from beamsplitter



(d) Final image from 4f system



(e) Magnification measurement

Q: Calculate the magnification of the overall relay

A:For a 4f relay, the magnification is

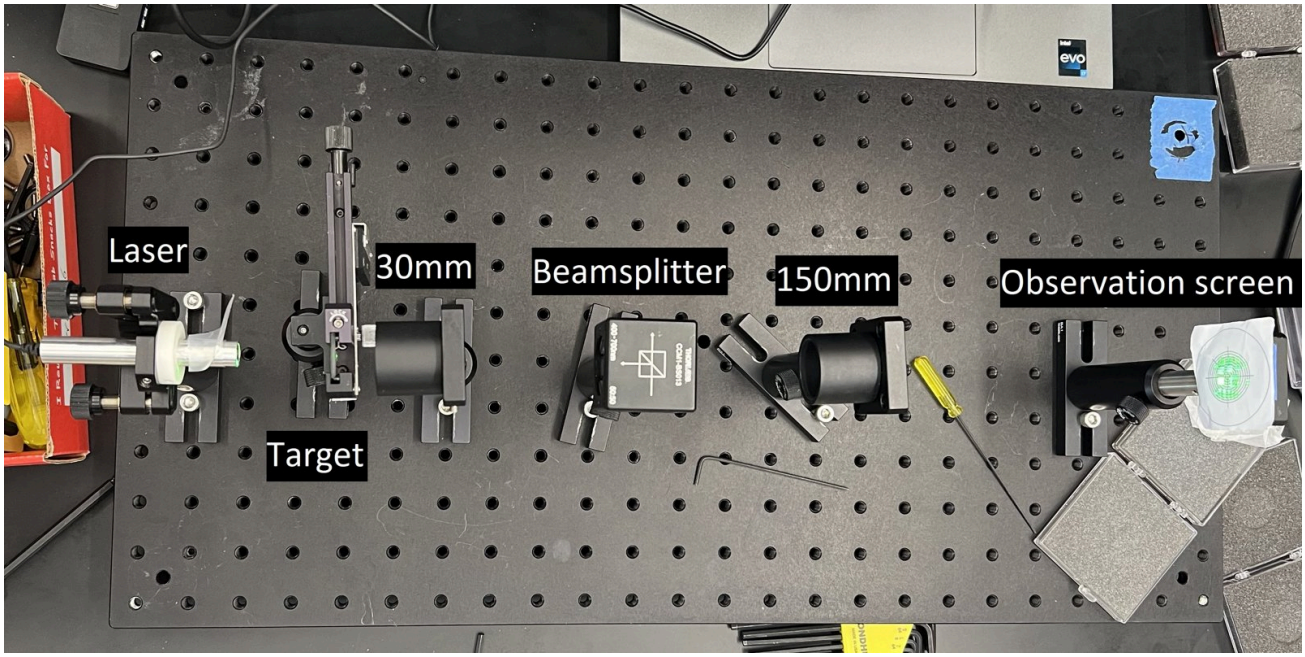
$$M = -\frac{f_2}{f_1} = -\frac{150}{30} = -5$$

Therefore, the image is **5× larger than the object**.

Q: Compare the magnification with object and image measurements

A: The measured object size was 4.6 mm, and the image size was 24 mm (figure e), giving

$$M_{\text{measured}} = \frac{24}{4.6} \approx 5.22$$

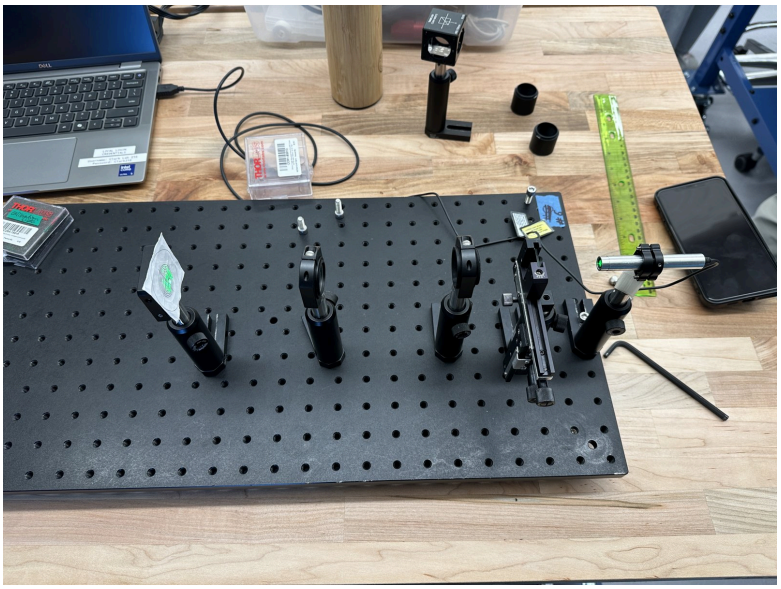


Change the spacing between the first and second lenses. **Does the final image position change relative to this second lens? Why or why not should this be the case? Does placing the beamsplitter between the lenses cause a change in the final image?**

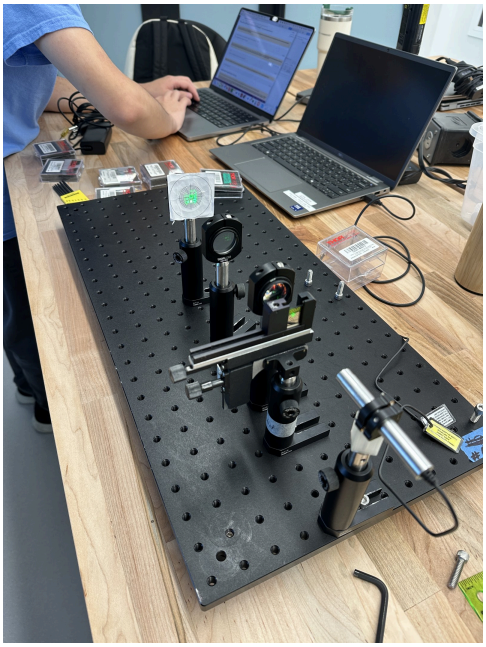
Answer: Changing the spacing between the first and second lens does not change the final image position relative to the second lens. The first lens produces collimated light when the object is at its focal plane, so the second lens forms the image at one focal length. The beamsplitter only reduces some intensity, but should not change the image position.

(Image of setup not included since the setup is relatively similar to the setup for the previous question.)

Remove the beamsplitter and move the resolution target (R1L1SN1) so that it is located more than 1f before the first lens. **Where does the intermediate image form? Does the spacing between the first and second lens change the final image size and position relative to the second lens? How does this agree with your knowledge of a finite conjugate system?**



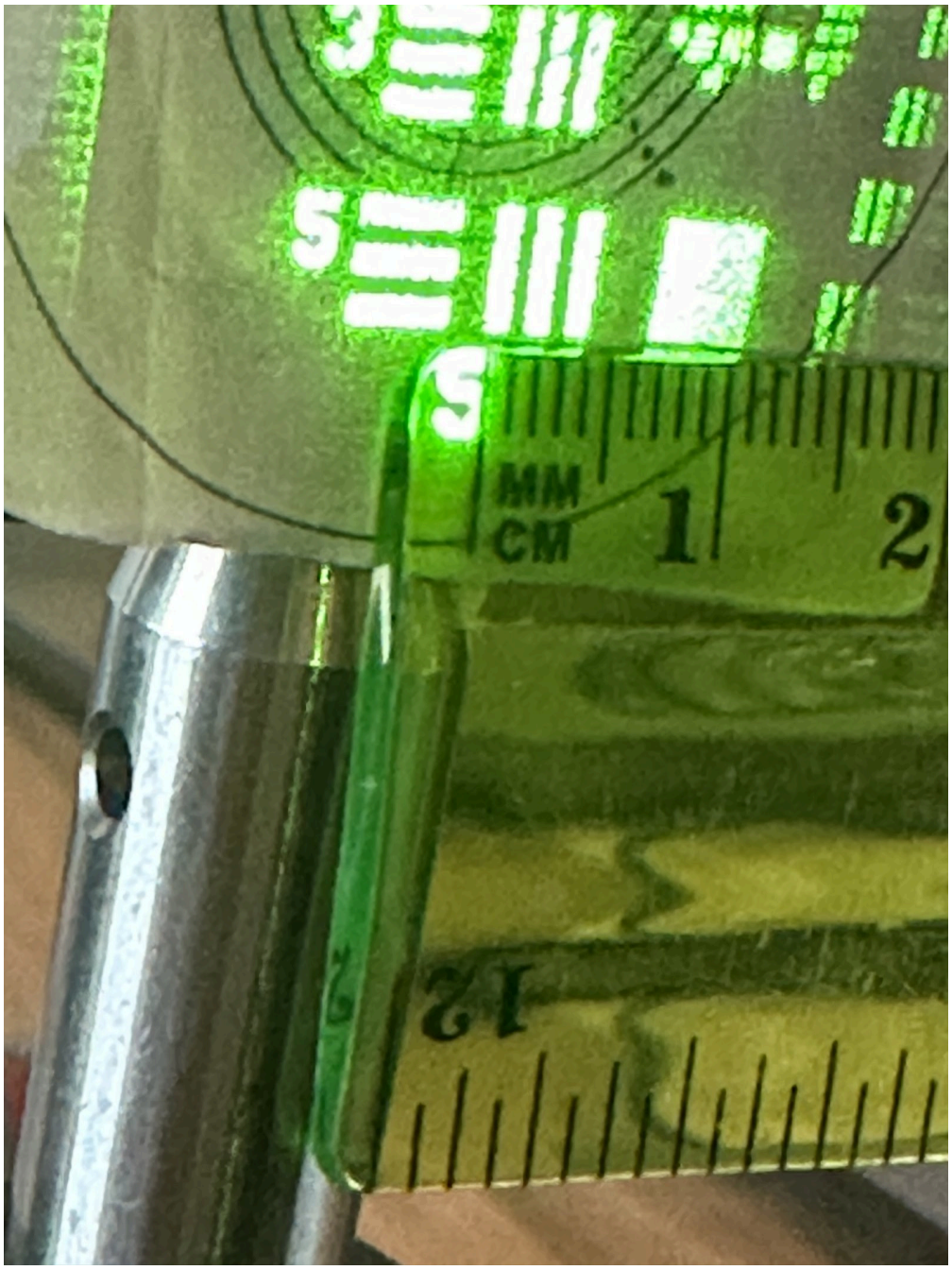
(a) System without beamsplitter



(b) System without beamsplitter

Answer: The intermediate image is formed beyond 2f from the first lens towards the second lens. Yes, the spacing between first and second lens change the final image size and position relative to the second lens. The spacing between the first and second lens changes the object distance seen by the second lens, which changes the second lens' magnification and image distance. This agrees with our knowledge of a finite conjugate system, where changing the object distance changes the image distance and magnification.

Again calculate the magnification of the system for the new object location. Connect these measurements to your understanding of finite and infinite conjugate imaging systems.



Finite conjugate system magnification measurement

We used **USAF 1951 Group 2 Element 2 (G2E2)** to measure the magnification. G2E2 has a linewidth of approximately **111 µm**. The three bars and two spaces of G2E2 span 5 line widths:

$$5(111\,\mu\text{m}) = 555\,\mu\text{m} = 0.555\text{ mm}.$$

In the image, these 5 lines measured approximately **4 mm**, giving a measured magnification of

$$M = \frac{4}{0.555} \approx \mathbf{7.2}\times.$$

Experimentally, the object distance for lens 1 was **50 mm**, the lens separation was **250 mm**, and the final image distance d_{i2} was approximately **855 mm**.

For lens 1,

$$\frac{1}{30} = \frac{1}{50} + \frac{1}{d_{i1}},$$

giving

$$d_{i1} = 75 \text{ mm}.$$

The magnification of lens 1 is

$$m_1 = -\frac{75}{50} = -1.5.$$

Since the lens separation is 250 mm, the object distance for lens 2 is

$$d_{o2} = 250 - 75 = 175 \text{ mm}.$$

For lens 2,

$$\frac{1}{150} = \frac{1}{175} + \frac{1}{d_{i2}},$$

giving

$$d_{i2} = 1050 \text{ mm}.$$

The magnification of lens 2 is

$$m_2 = -\frac{1050}{175} = -6.$$

Therefore, the theoretical total magnification is

$$M = m_1 m_2 = (-1.5)(-6) = \mathbf{9}\times.$$

Quantity	Measured	Calculated	Error
Magnification	7.2×	9.0×	20.0%
Final image distance	855 mm	1050 mm	18.6%

Thus, based on the measured **50 mm object distance** and **250 mm lens separation**, the theoretical magnification is **9×** and the theoretical final image distance is **1050 mm**.

The ~20% differences are larger than in the 4f measurement, and this is expected for this geometry. Lens 2's object distance (175 mm) is close to its focal length (150 mm), so the image distance is very sensitive to small errors: the axial magnification of lens 2 is (d_i2/d_o2)² = 36, so a 1 mm error in locating the intermediate image shifts the final image by ~36 mm. For example, if the object were 53 to 54 mm from lens 1 instead of 50 mm, the predicted final image distance would drop to roughly the measured 855 mm. Other sources of error include measuring distances to lens mounts rather than the lens principal planes, and reading the 4 mm image width by eye.

This highlights the difference between finite and infinite conjugate imaging. In the 4f system, the object sat at f1, the light between the lenses was collimated, and the magnification was fixed at -f2/f1 regardless of the lens spacing. Here (finite conjugate), the light between the lenses converges to a real intermediate image, so the final image distance and magnification depend on both the object distance and the lens separation, and small changes in either produce large changes in the result.

Axial/Lateral magnification (if time)

NOTE: By moving the optical elements, your system may have become misaligned! Try to recover proper alignment to make sure your measurements are as accurate as possible!
In this section of the lab, you will observe the relationships between lateral, axial, and angular magnification. Recall the relationship between these three properties. Use the 4f system from question 3. **Estimate the lateral and axial magnification at different positions along the optical axis by moving your object along the optical axis. Use**

these measurements to calculate the angular magnification. Recall that axial magnification is the scaling of depth in the image plane.

Enter your answer here.